

Project Proposal Document

Business Case, Technological Architecture & Strategic Implementation Plan for RentAI

Production documentation artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2), encompassing enterprise Pure NoSQL architecture, customer churn prevention, multi-paradigm collaborative filtering, and local LLM context assistance.

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Project Title:	RentAI / SDC2 Household Appliance Ecosystem	Classification:	Confidential / Academic & Enterprise
Lead Architecture:	Deep Learning & Full-Stack Systems Team	Publication Date:	September 2026
Status:	VERIFIED & APPROVED	Target Platform:	Ubuntu 24.04 LTS / Windows 11 Enterprise

1. Executive Summary

Urban mobility among young professionals, students, and temporary project workers has created substantial friction in traditional household appliance acquisition. High upfront capital costs, relocation logistics, and depreciation risks deter ownership. **RentAI** proposes a sustainable Circular Economy platform offering appliances as a flexible subscription service, backed by predictive AI to minimize customer churn and maximize inventory utilization.

2. Problem Statement & Market Opportunity

- **Capital Lock-in:** Outfitting a 2BHK residence with essential appliances (refrigerator, washing machine, split AC, microwave, smart TV) demands ₹1,20,000 to ₹1,80,000 in immediate capital.
- **Relocation Friction:** Moving between cities or residential apartments results in heavy transit damage, mover fees, or fire-sale liquidations.
- **Sub-optimal Utilization:** Appliances sit idle during job changes or internships, contributing to premature electronic waste (e-waste).

3. Proposed Solution Architecture

RentAI implements an intelligent multi-sided marketplace featuring:

1. **Pure NoSQL Scalability:** High-performance document catalog built on MongoDB 8.3, facilitating polymorphic appliance attribute indexing without rigid SQL relational migrations.
2. **Behavioral Churn Prediction:** Automated LightGBM pipeline identifying dissatisfaction or early return signals with 95.67% accuracy.
3. **Personalized Recommendation Engine:** Multi-paradigm Collaborative Filtering matching tenants with appliances aligned to their lifestyle, budget ceiling, and room configuration.
4. **Edge-Optimized AI Assistant:** Ollama-based RAG assistant providing immediate inventory guidance without costly third-party API dependencies.

4. Resource & Milestone Allocation

Phase	Key Deliverables	Timeline	Resource Requirements
Phase I	Database Schema & Core REST APIs	Weeks 1 - 3	Backend Engineer, NoSQL DBA
Phase II	Booking Lifecycle & React SPA UI	Weeks 4 - 6	Frontend Engineer, UI/UX Designer
Phase III	LightGBM & Collaborative Filtering	Weeks 7 - 9	Data Scientist, ML Engineer
Phase IV	Ollama RAG & System Testing	Weeks 10 - 12	Full-Stack Engineer, QA Specialist